



4896 - AI-Enabled Modeling of Instrument Systematics for Exoplanet Time-Series Observations

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Exoplanet observations require extremely precise measurements to detect atmospheric signals, particularly on terrestrial targets. At these precisions, telescope systematics such as ramps, thermal fluctuations, voltage activations of observatory heaters, etc. are present as additional noise sources in the data. However, telescopes like JWST have hundreds of instrument sensors tracking the temperatures and voltages throughout the observatory systems. This archival proposal seeks to use machine learning tools to characterize the systematics in exoplanet time series data for the most commonly used JWST Exoplanet time series modes. MIRI/LRS has the greatest number of integrations, while NIRSpec/G395H has the greatest number of individual programs. We will first use a random forest to conduct feature engineering tests and determine the basis mnemonics that drive the red noise. These will then be used to train the random forest to predict the systematics signal present in reduced white light curves. We will provide the model and a description of its implementation for the community to use. We will also incorporate new software into the appropriate

reduction stage of the Eureka! pipeline. This proposal is pioneering work and expected to have a high impact on all exoplanet time series analyses by improving the precision of dozens of exoplanet programs.

OBSERVING DESCRIPTION